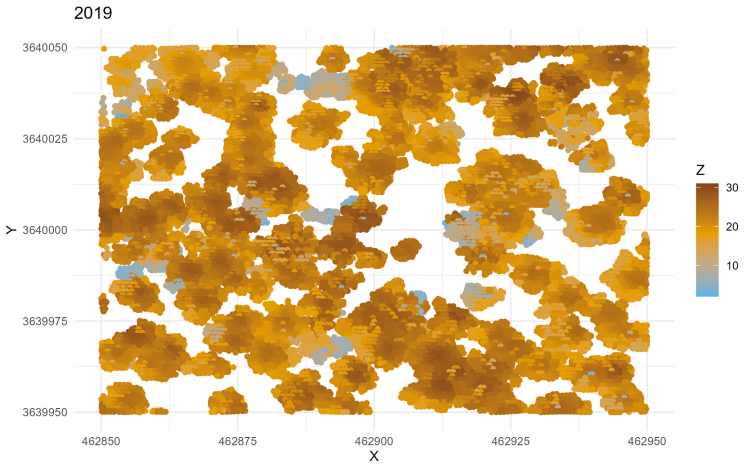
Final Report

Counting Trees

Team :
Fred Jiagho
Rajvi Shah
Oscar Goudon de Lalande
Mouad Leachouri
Chihab Majdolin

Supervisors : Konstantinos Florakis and Gilles Faÿ



Pole No. 15

Mathematical Modeling of Complex Systems

May 2025

Contents

1 Abstract **3**

2 Introduction **3**

3 State of Art **4**

4 Theoretical Background **4**

4.1 Spatial Point Processes 4

4.1.1 Poisson Point Process 5

4.1.2 Thomas Point Process 6

4.1.3 Superposition of Point Processes 8

4.2 Pair Correlation Function 8

4.3 Spectral Density of a Point Process 10

4.4 Bartlett's Periodogram 10

5 Parameter Estimation **12**

5.1 Methodology 12

5.2 Application to one Thomas process 12

5.2.1 Using the Pair Correlation Function (PCF) 12

5.2.2 Using the Spectral Density 13

5.2.3 Tapering 14

5.3 Application to a superposition of two independent Thomas Processes 21

5.3.1 Assumptions 21

5.3.2 Estimation using the PCF 22

5.3.3 Estimation using the Fourier Transform 22

6 Conclusion and Perspectives **23**

6.1 Conclusion 23

6.2 Perspectives 24

Appendix **27**

A Intensity of a superposition of point processes **27**

B Pair correlation function of a superposition of point processes **27**

C Fourier Transform of a superposition of point processes **27**

D Expression of the individual intensities for a superposition **28**

E PCF of a superposition under hypotheses 5.3.1 **28**

F Intensity estimation **29**

1. Abstract

Forests are incredibly important for our environment, supporting biodiversity and providing valuable resources. To manage them effectively and understand their ecology, we need to know how many trees there are and how they're spread out. Traditional ways of counting trees in the field are often slow and expensive. While LiDAR technology gives us detailed 3D scans of forests, getting useful information like tree counts from this raw "point cloud" data is a real challenge because it loses some of the original structural information.

Our project tackles this by using a statistical model called a superposition of Thomas processes. This helps us reconstruct the hidden spatial patterns of trees from the LiDAR data. With this model, we can accurately count trees and analyze how they're clustered, even pinpointing areas with different tree densities. We're using advanced methods, harnessing the Pair Correlation Function (PCF) and Fourier Transform, to estimate these model parameters. We also use periodogram plotting and aim to significantly improve the accuracy of our estimates for Thomas processes. To make sure our results are robust, we measure accuracy using metrics like Absolute Percentage Error and reduce tricky edge effects by using spectral tapering, which reduces the variance of our estimators. We aim to test our approach with real LiDAR data from forests.

2. Introduction

Forests play a crucial role in regulating our environment, supporting an incredible diversity of life, and providing resources we can sustainably manage [1, 2, 3]. Knowing precisely what types of trees are present and how many of them there are is vital for measuring carbon storage, guiding forestry practices, and advancing ecological research [4, 5].

Historically, forest inventories relied on hands-on field surveys, which are often time-consuming and costly. LiDAR (Light Detection and Ranging) offers a promising alternative, providing incredibly detailed 3D point clouds of wooded areas. However, extracting practical ecological insights like the number and exact distribution of trees from this raw LiDAR data can be computationally tricky [6, 7, 8]. The raw LiDAR output typically comes as a spatial point process—essentially just a collection of points in space. This representation can unfortunately obscure the underlying structural patterns of trees and how they're clustered, making it hard to directly figure out tree counts and their spatial organization.

Our current work aims to solve this challenge using tools from stochastic geometry. We propose to model how trees are distributed in space using spatial point processes, particularly clustered ones, mainly Thomas processes. We then estimate their parameters using spectral and non-spectral methods. This approach lets us piece back together the lost structural information from the LiDAR point process data, leading to more accurate tree counts and a much clearer understanding of their spatial layout.

3. State of Art

Currently, most methods for getting individual tree information from LiDAR data rely on segmentation techniques [9, 10, 11]. These techniques seek to isolate the crowns, or even the trunks, of the trees from the point cloud, which then allows them to be counted and characterized. While these methods work well for trees that are clearly separated and stand out, they run into significant problems with understory trees, crowns that heavily overlap, or smaller, suppressed trees. These often get poorly segmented or missed entirely, leading to an underestimation of the actual tree density [7]. This limitation comes from how difficult it is to pick out individual tree structures within dense or complex forest canopies when you're just looking at geometric shapes.

Our project takes a different approach to overcome this limitation. Instead of focusing on perfectly precise individual tree positions (which can be hard to get for hidden trees), we prioritize getting an accurate overall tree count and understanding their spatial clustering. Rather than trying to segment every single tree, we model the realization of LiDAR on forest as a superposition of spatial point processes, specifically Thomas processes. This statistical modeling lets us infer the underlying clustering patterns and the overall tree density directly from the LiDAR point process data, even when individual trees aren't perfectly clear. By doing this, we aim to provide a more robust estimate of how many trees are present, especially in challenging forest environments where traditional segmentation methods fall short. Essentially, we're trading off exact positional accuracy for improved counting and a better understanding of the spatial distribution. Our work is heavily influenced by recent breakthroughs in analyzing spatial point processes using spectral methods, building upon foundational research like "What is the Fourier Transform of a Spatial Point Process" by Rajala et al. [12].

4. Theoretical Background

Our approach is principally informed by [13].

4.1. Spatial Point Processes

A **spatial point process** (more precisely, a *2D* spatial point process) is a probabilistic model that describes the random distribution of points in a set $S \subset \mathbb{R}^2$. A realization of the process is an *at most countable* set of points $\{x_1, x_2, \dots\} \subset S$. [14]

A spatial point process can be described in two ways:

- Using a **random counting measure** Φ. For any Borel set A, we denote by $\Phi(A)$ the

number of points falling in A.[1]

- Using a **random set** Φ such that, for a certain outcome ω, $\Phi(\omega)$ is the set of points for that outcome. A set is a convenient object to describe this, as ordering is irrelevant.[2]

4.1.1 Poisson Point Process

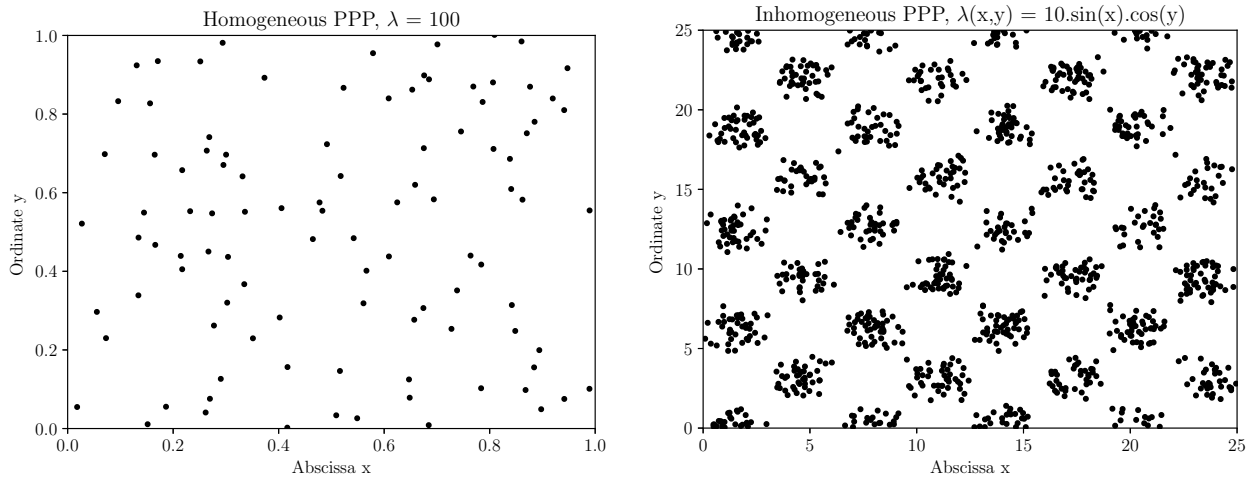
Definition (Poisson Point Process). A *Poisson point process* (PPP) on $\mathbb{R}^2$ is a random collection of points characterized by an intensity function $\lambda(\cdot) : \mathbb{R}^2 \to [0, \infty)$ that governs the expected density of points in space. We distinguish two main cases:

- **Homogeneous Poisson Point Process (See Fig. 1a):** If the intensity function is constant, i.e., $\lambda(x) = \lambda > 0$ for all $x \in \mathbb{R}^2$, the process is called *homogeneous* or *stationary*. It satisfies the following properties:
  - **Intensity:** The expected number of points per unit area is constant, equal to λ.
  - **Poisson distribution of counts:** For any bounded Borel set $A \subset \mathbb{R}^2$, the number of points $\Phi(A)$ in A follows a Poisson distribution of parameter $\lambda|A|$, where $|A|$ is the Lebesgue measure (area) of A:
  $$\Phi(A) \sim \text{Poisson}(\lambda|A|).$$
  - **Independence:** For any collection of disjoint bounded Borel sets $A_1, A_2, \ldots, A_n$, the random variables $\Phi(A_1), \Phi(A_2), \ldots, \Phi(A_n)$ are mutually independent.
- **Non-Homogeneous (Inhomogeneous) Poisson Point Process(See Fig. 1b):** If the intensity function varies over space, i.e., $\lambda(x) \geq 0$ is a measurable function that may depend on $x \in \mathbb{R}^2$, the process is called *non-homogeneous*. It satisfies the following:
  - **Intensity function:** $\lambda(x)$ describes the expected local density of points at x.
  - **Poisson distribution of counts:** For any bounded Borel set $A \subset \mathbb{R}^2$, the number of points in A, $\Phi(A)$, follows a Poisson distribution with parameter given by the integral of the intensity over A:
  $$\Phi(A) \sim \text{Poisson}\left(\int_A \lambda(x)\, dx\right).$$
  - **Independence:** For any collection of disjoint bounded Borel sets $A_1, A_2, \ldots, A_n$, the counts $\Phi(A_1), \Phi(A_2), \ldots, \Phi(A_n)$ are independent random variables.

The Poisson point process, in both homogeneous and non-homogeneous forms, models *complete spatial randomness* (CSR) in the sense that points occur independently over space according to the intensity function. It inherently assumes no interaction between points—no clustering or inhibition effects are present.

[1] It's a *random* measure because it depends on the point process, hence its randomness.

[2] The reader should keep in mind the abuse of notation. Both concepts (random set and random measure) are denoted Φ.



(a) Homogeneous Poisson Process

(b) Inhomogeneous Poisson Process

Figure 1: Homogeneous vs. Inhomogeneous Poisson Process

4.1.2 Thomas Point Process

Definition (Thomas Process). The Thomas process is a *clustered spatial point process* defined on $\mathbb{R}^2$, and is constructed as follows:

- **Parent process:** Parent points, i.e., centers of the clusters (note that parent points are *not* included in the final point process) form a homogeneous Poisson point process with intensity λ_p.
- **Offspring count:** Each parent point produces a random number of offspring points, independently, following a Poisson distribution with mean ν.
- **Offspring location:** Given a parent located at position $\mathbf{c} \in \mathbb{R}^2$, each offspring is independently placed according to a bivariate normal distribution centered at $\mathbf{c}$, with density:

$$f(\mathbf{x} \mid \mathbf{c}) = \frac{1}{2\pi\sqrt{\det(\mathbf{\Sigma})}} \exp\left(-\frac{1}{2}(\mathbf{x}-\mathbf{c})^\top \mathbf{\Sigma}^{-1}(\mathbf{x}-\mathbf{c})\right)$$

We assume that the covariance matrix $\mathbf{\Sigma}$ is isotropic and scalar, i.e., $\mathbf{\Sigma} = \sigma^2 \mathbf{I}_2$, where $\mathbf{I}_2$ is the 2×2 identity matrix.

The total intensity of the Thomas process is given by:

$$\lambda = \lambda_p \, \nu.[13]$$

This is the main model we will use because, as can be seen in Fig. 2, it models very well the pattern produced by a LiDAR (Fig. 3).

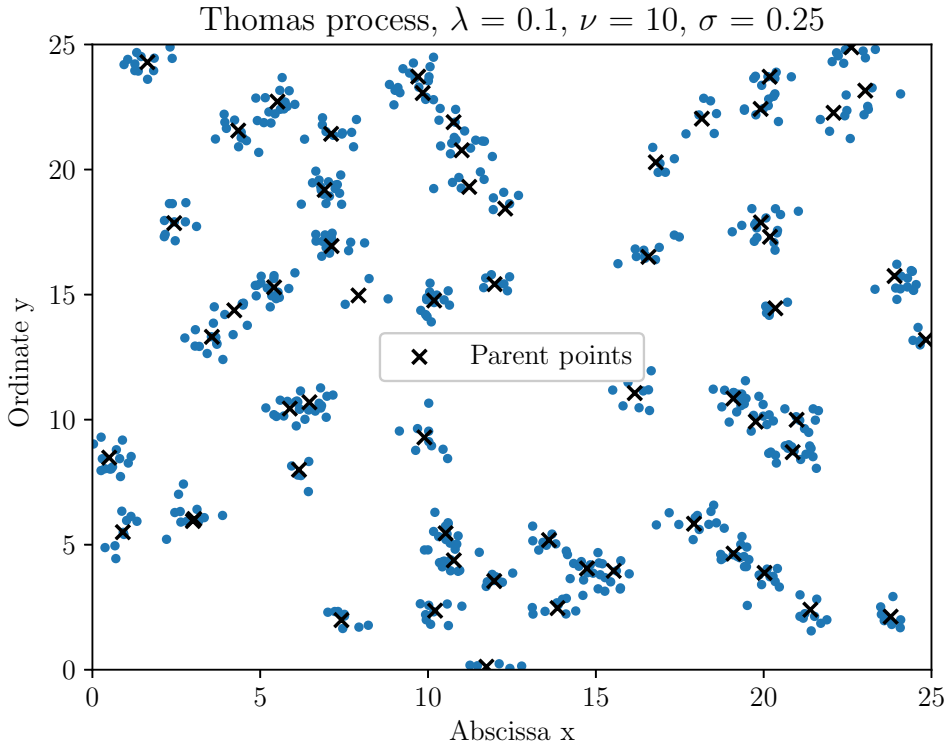


Figure 2: Realisation of a Thomas process

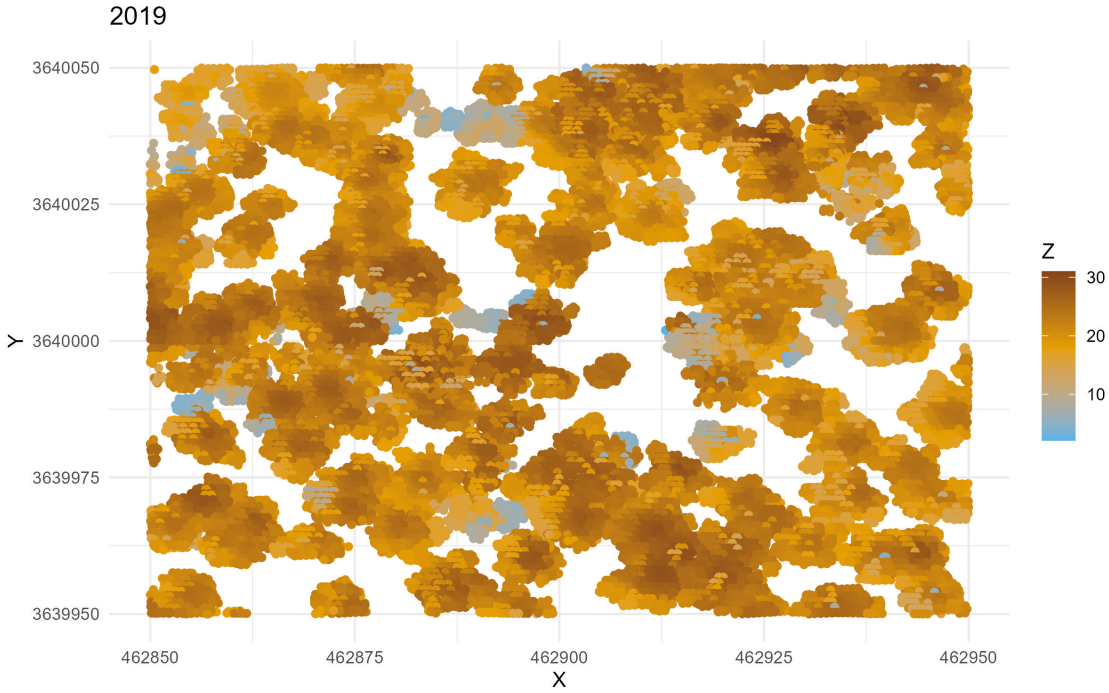


Figure 3: LiDAR scan of a forest, 2019

4.1.3 Superposition of Point Processes

Definition (Superposition of Point Processes). Let $\Phi_1, \Phi_2, \ldots, \Phi_n$ be n spatial point processes defined on $\mathbb{R}^2$. The *superposition* of these processes is the point process

$$\Phi = \bigcup_{i=1}^{n} \Phi_i,$$

i.e., Φ consists of all the points from each Φ_i. If the processes $\Phi_1, \ldots, \Phi_n$ are independent, the superposition Φ is also a point process, often denoted as the independent superposition (see Fig. 4).

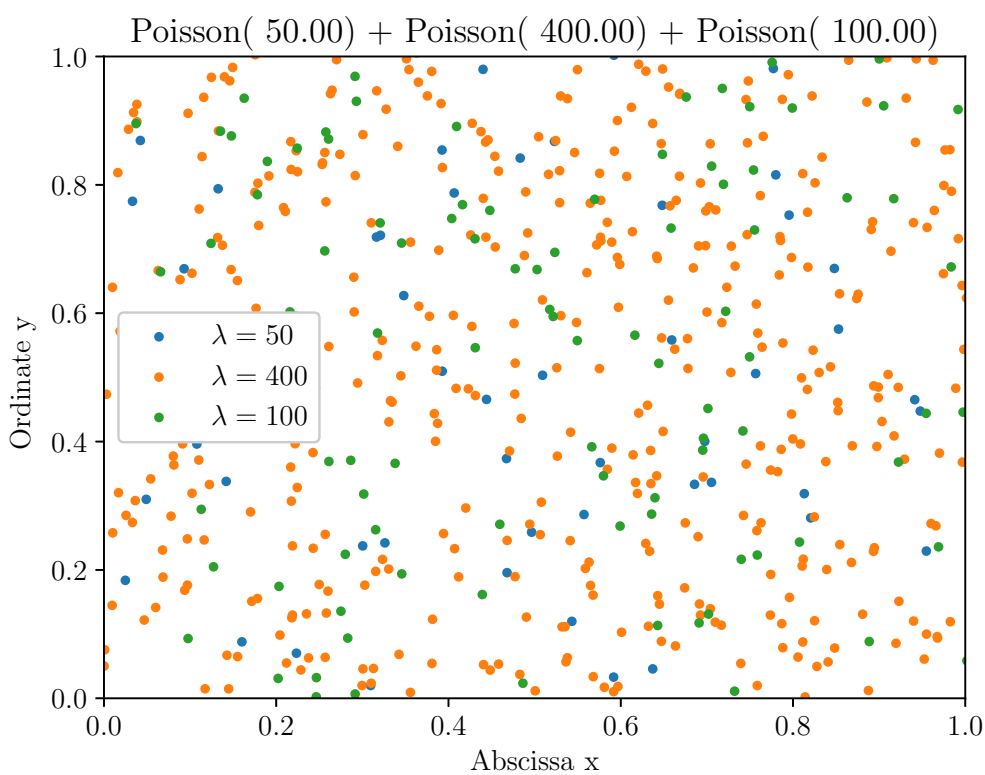


Figure 4: Superposition of three Poisson processes

4.2. Pair Correlation Function

Intuitively, one can think of the **pair correlation function** g of a stationary and isotropic[3] point process as describing how the density of pairs of points depends on their separation distance r [15].

[3] A stationary isotropic point process is invariant under translations and rotations, meaning its statistical properties are the same everywhere and in every direction.

For a stationary isotropic process, it can be defined as:

$$g(x, y) = g(\|x - y\|) = g(r) = \frac{\rho^{(2)}(r)}{\lambda^2},$$

for any two points x and y, where $\rho^{(2)}(r)$ is the second-order product density,[4] and λ is the intensity of the process.

For a Thomas process, the pair correlation function takes the explicit form:[15]

$$g(r) = 1 + \frac{1}{4\pi\lambda_p\sigma^2} \exp\left(-\frac{r^2}{4\sigma^2}\right).$$

This expression reveals the clustering behavior of the process: $g(r) > 1$ for small r, and $g(r) \to 1$ as $r \to \infty$, indicating that point interactions diminish at large distances.

Let's note that it's possible to estimate the pair correlation function of a point process using the Kernel density estimation.[15] Fig.5 shows the estimator for a superposition of two Thomas processes.

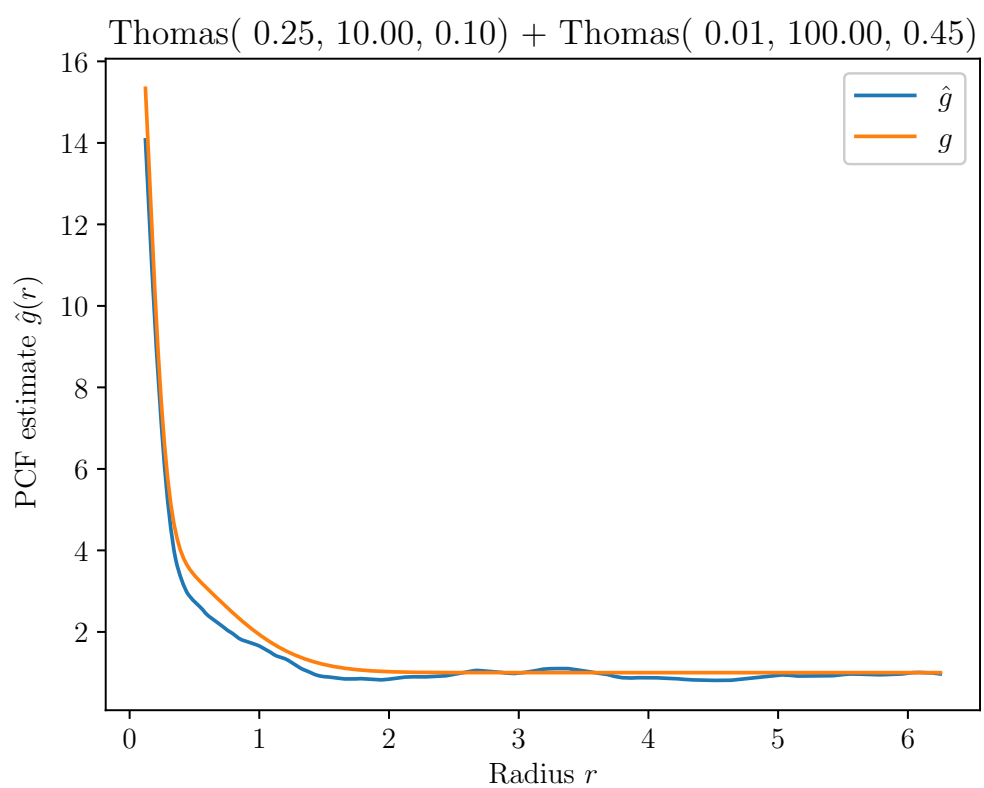


Figure 5: Theoretical and estimated pair correlation function

[4]See [15] for the definition of $\rho^{(2)}$ and further detail.

4.3. Spectral Density of a Point Process

Let Φ be a stationary isotropic spatial point process with intensity λ. The complete covariance of Φ is defined as:

$$\gamma(z) = \lambda\delta(z) + \rho^{(2)}(z) - \lambda^2.$$

The spectral density function of Φ is the Fourier transform of the complete covariance:[12]

$$\forall k \in \mathbb{R}^2 : \; f(k) = \mathcal{F}[\gamma](k) = \lambda + \int_{\mathbb{R}^2} e^{-2\pi i k \cdot z} \left[\rho^{(2)}(z) - \lambda^2\right] dz$$

In particular, for a Thomas point process of parameters (λ_p, ν, σ), the Fourier Transform is given by:[12]

$$\forall k \in \mathbb{R}^2 : \; f(k) = \lambda_p \nu \left(1 + \nu \exp\left(-4\pi^2 \|k\|_2^2 \sigma^2\right)\right)$$

This transformation allows us to analyze spatial structures in the frequency domain.

4.4. Bartlett's Periodogram

Given an observation window $W \subset \mathbb{R}^2$ and a realization of points $X = \{x_1, \ldots, x_n\}$, the **Bartlett periodogram** is an estimator of the spectral density function, given by:[12]

$$I_0(k) = \frac{1}{|W|} \sum_{x,y \in X \cap W} e^{-2\pi i k \cdot (x-y)}$$

It is a biased estimator due to edge effects and aliasing but is foundational in spectral estimation. Fig.6 shows the Bartlett periodogram for a Thomas point process with parameters $(\lambda_p = 0.25, \nu = 10, \sigma^2 = 0.1^2)$

Let's note that all these definitions that we introduced in the case of a simple point process (Intensity, Pair correlation function, Fourier Transform, Bartlett's periodogram) can also be used in the case of a superposition of independent point processes (See Appendix A, Appendix B, Appendix C for more details). And as the estimators we use are non-parametric, we can easily estimate these statistics for a superposition of independent point processes.

For example, Fig.7 shows the Barlett periodogram for a superposition of two Thomas processes with parameters $(0.25, 10, 0.1^2)$ and $(0.01, 25, 0.45^2)$.

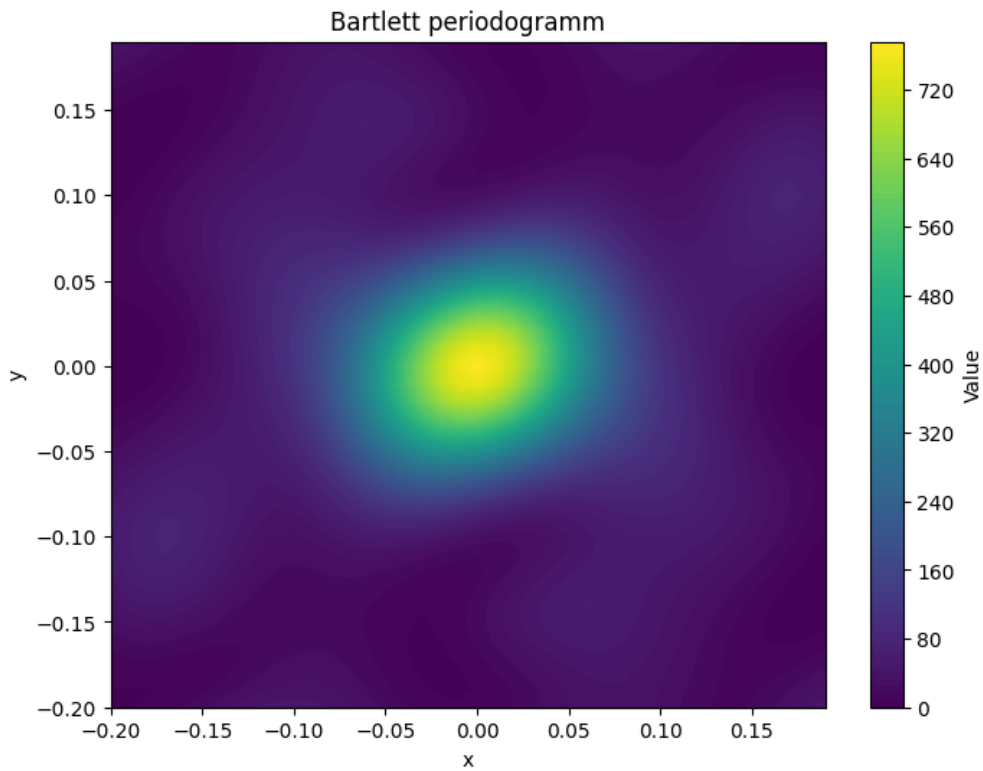


Figure 6: Barlett Periodogram for a Thomas process with parameters $(0.25, 10, 0.1^2)$

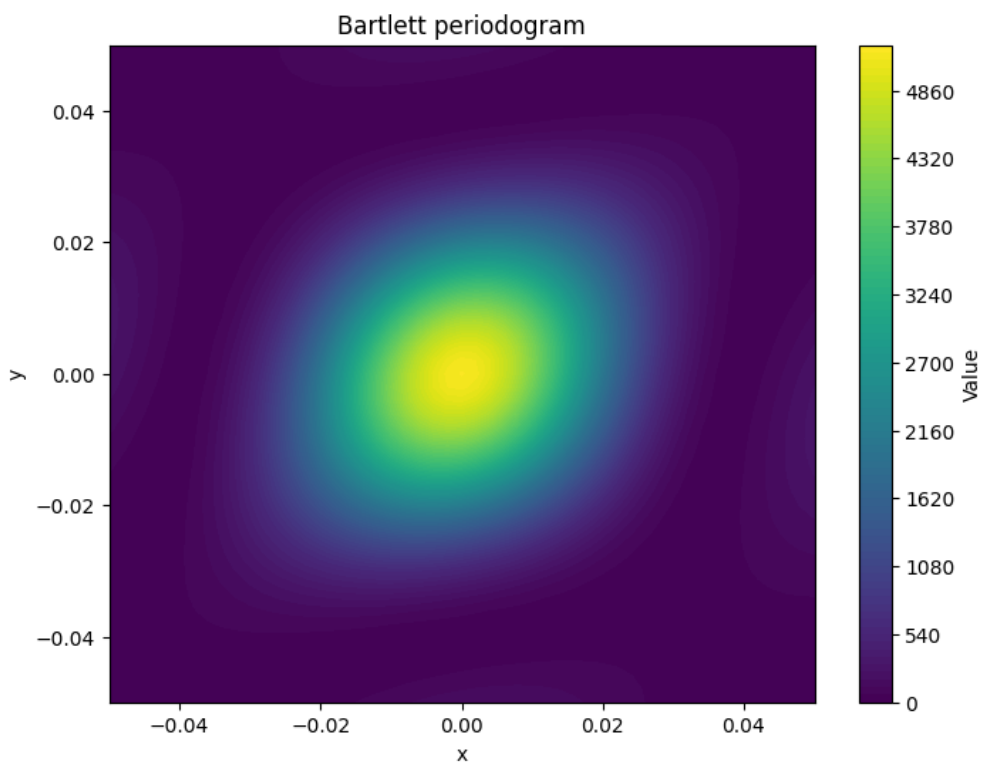


Figure 7: Barlett's Periodogram for a superposition of Thomas point processes

5. Parameter Estimation

5.1. Methodology

We aim to estimate the parameters of a superposition of Thomas process. For this, we use the method described in [15].

Let θ denote the vector of parameters to estimate. We consider a theoretical quantity T_θ (in our case, the PCF g or the spectral density f) that depends on θ, and we denote by $\hat{T}$ its non-parametric empirical estimate based on the available data.

The parameter estimate $\hat{\theta}$ is obtained by minimizing a contrast function, defined as the L^2 distance between T_θ and $\hat{T}$:

$$\Delta(\theta) = \int_{r_1}^{r_2} \left(T_\theta(r) - \hat{T}(r) \right)^2 dr.$$

Ideally, we would integrate over the entire positive real axis, i.e., choose $r_1 = 0$ and $r_2 = +\infty$. However, in practice, this leads to divergence because the estimator $\hat{T}$ becomes unstable near 0 and unbounded at infinity. To avoid this, we restrict the integration domain to $[r_1, r_2] := [\varepsilon, R]$, where ε is a small positive number and R is a sufficiently large number, typically of the same order of magnitude as the size of the observation window.

To compute this integral, we use a discrete sum over K values of r in $[\varepsilon, R]$:

$$\hat{\Delta}(\theta) = \sum_{i=1}^{K} \frac{R - \varepsilon}{K} \left(T_\theta(r_i) - \hat{T}(r_i) \right)^2,$$

where $\varepsilon = r_1 < r_2 < \cdots < r_K = R$.

5.2. Application to one Thomas process

5.2.1 Using the Pair Correlation Function (PCF)

We apply the previously described estimation procedure using the pair correlation function g of a Thomas process with true parameters given by the vactor $\theta = (0.3,\ 10,\ 0.25^2)$.

It is important to note that for a Thomas process, the pair correlation function depends only on the parent intensity λ_p and the dispersion parameter σ, but not on the expected number of offspring ν. Therefore, we first estimate λ_p and σ by minimizing the contrast based on the

PCF. And, given an empirical estimate of the global intensity λ,[5] denoted by

$$\hat{\lambda} = \frac{\Phi(B)}{|B|},$$

where $|B|$ is the area of the observation window, and using the identity $\lambda = \lambda_p \nu$,[6] we can estimate ν using:

$$\hat{\nu} = \frac{\hat{\lambda}}{\hat{\lambda}_p}.$$

To assess the quality of our estimation, we compute the **absolute percentage error**. For a single parameter θ_i, this is given by:

$$\text{Error}(\theta_i) = \frac{|\theta_i - \hat{\theta}_i|}{|\theta_i|}.$$

Table 1 summarizes the results obtained for a simulated Thomas process.

	$\boldsymbol{\lambda_p}$	$\boldsymbol{\nu}$	$\boldsymbol{\sigma}$
True values	0.300	10.000	0.250
Estimated values	0.298	10.137	0.254
Percentage error	1.4%	1.6%	1.4%

Table 1: Parameter estimation for one Thomas process, using the PCF

As shown in the table, the estimated values are very close to the true values. This confirms the effectiveness of the estimation procedure when using the pair correlation function as a contrast function.

5.2.2 Using the Spectral Density

We now aim to apply the same estimation methodology using the **Fourier Transform** of the point process, along with its empirical estimator—the **Bartlett periodogram**. However, this approach presents a major issue related to parameter identifiability.

The theoretical Fourier Transform of a Thomas process is given by:[12]

$$f(k) = \lambda \left(1 + \nu\, e^{-4\pi^2 \|k\|^2 \sigma^2}\right) = \lambda_p \nu \left(1 + \nu\, e^{-4\pi^2 \|k\|^2 \sigma^2}\right).$$

[5] Appendix F summarizes our work on the following, as well as another more sophisticated estimator, more suited for estimating second order characteristics

[6] The global intensity (points per unit area) of a Thomas process is simply the sum of the contributions of each cluster. Since each cluster has ν offsprings on average, the global intensity is $\nu \cdot \lambda_p$

Because $\lambda = \lambda_p \nu$, the contrast depends on the product $\lambda_p \nu$, rather than on each parameter individually. As a result, infinitely many combinations of λ_p and ν can yield the same value of λ, and therefore, the same value of the contrast function. This makes the model **non-identifiable** when using the Fourier Transform directly. It's good to notice that the same problem exists for the pcf approach but the pcf() function of the spatstat handles it directly.

To overcome this issue, we replace λ with its empirical estimate $\hat{\lambda}$ (introduced in 5.2.1). Then, we estimate ν and σ by minimizing the contrast based on this fixed $\hat{\lambda}$. Finally, we recover an estimate of λ_p via:

$$\hat{\lambda}_p = \frac{\hat{\lambda}}{\hat{\nu}}.$$

	$\boldsymbol{\lambda_p}$	$\boldsymbol{\nu}$	$\boldsymbol{\sigma}$
True values	0.300	10.000	0.250
Estimated values	0.309	10.251	0.255
Percentage error	3.0%	2.5%	2.0%

Table 2: Parameter estimation for one Thomas process, using the FT

As shown in Table 2, the estimated values are once again close to the true values. Despite the identifiability issue, the workaround provides good approximations, suggesting that the estimation procedure remains effective under this modification.

5.2.3 Tapering

Theoretical concept In order to reduce spectral leakage and improve the estimation of second-order properties of a spatial point process, *tapering functions* are introduced into the Bartlett periodogram estimator. This approach is adapted from classical spectral estimation methods in time series, where windowing techniques are commonly used to control bias and variance.[12]

The tapered periodogram used in spectral analysis of spatial point patterns is derived from the general Fourier transform of the tapered empirical measure. For a generic tapering function $h : \mathbb{R}^2 \to \mathbb{C}$, the tapered periodogram is defined as in Equation (14) of [12]:

$$I_h(k) := |J_h(k)|^2 = \sum_{x \in X \cap B} h(x)h^*(x) + \sum_{\substack{x,y \in X \cap B \\ x \neq y}} h(x)h^*(y)e^{-2\pi i k \cdot (x-y)},$$

where X is the observed point pattern, B is a bounded subset of the observation window, and $k \in \mathbb{R}^2$ is the frequency vector.

When the tapering function is *isotropic*, i.e., $h(x) = h_I(\|x\|)$, the expression simplifies via radial integration and the use of the Bessel function of the first kind J_0, leading to the

isotropic form (Equation (17) from [12]):

$$\bar{I}_h(\|k\|) \equiv \hat{\lambda} + \sum_{\substack{x,y \in X \cap B \\ x \neq y}} h_I(\|x-y\|) J_0(2\pi \|k\| \cdot \|x-y\|),$$

where $\hat{\lambda}$ is the empirical intensity (5.2.1).

A particular case of interest arises when the tapering function h is chosen as the normalized indicator of B:

$$h_0(z) = |B|^{-1/2} \mathbf{1}_{\{z \in B\}}.$$

Substituting into the general definition gives:

$$I_0(k) \equiv \hat{\lambda} + |B|^{-1} \sum_{\substack{x,y \in X \cap B \\ x \neq y}} e^{-2\pi i k \cdot (x-y)}, \quad k \in \mathbb{R}^2.$$

Tapering acts as a smoothing operator (see Fig. 8) on the spectral estimate, with its Fourier transform $\mathcal{F}[h]$ playing a crucial role in the bias and variance of the periodogram. Specifically, in their paper, Rajala shows that the expectation of the tapered periodogram is:

$$\mathbb{E}[\hat{f}_h(k)] = f(k) + \lambda^2 |\mathcal{F}[h](k)|^2 + o(1),$$

indicating a bias term that depends on the shape of $\mathcal{F}[h]$. Removing this bias gives rise to a *debiased periodogram*.

Parameter estimation with L^2-normalized isotropic tapers The goal of tepering is to see if we could have more accurate parameter estimation, using differents tapering functions. As a first approach, we use L^2-normalized isotropic tapering functions, exploiting the following estimator:

$$\bar{I}_h(\|k\|) \equiv \hat{\lambda} + \sum_{\substack{x,y \in X \cap B \\ x \neq y}} h_I(\|x-y\|) J_0(2\pi \|k\| \cdot \|x-y\|).$$

According to [12], an isotropic taper function h must be:

- **Isotropic**, i.e., $h(x) = h(\|x\|)$.
- **Square integrable** (finite energy): $\int_{\mathbb{R}^2} |h(x)|^2 dx < \infty$.
- Preferably, **compactly supported** or decay rapidly at the boundary of W to reduce leakage.

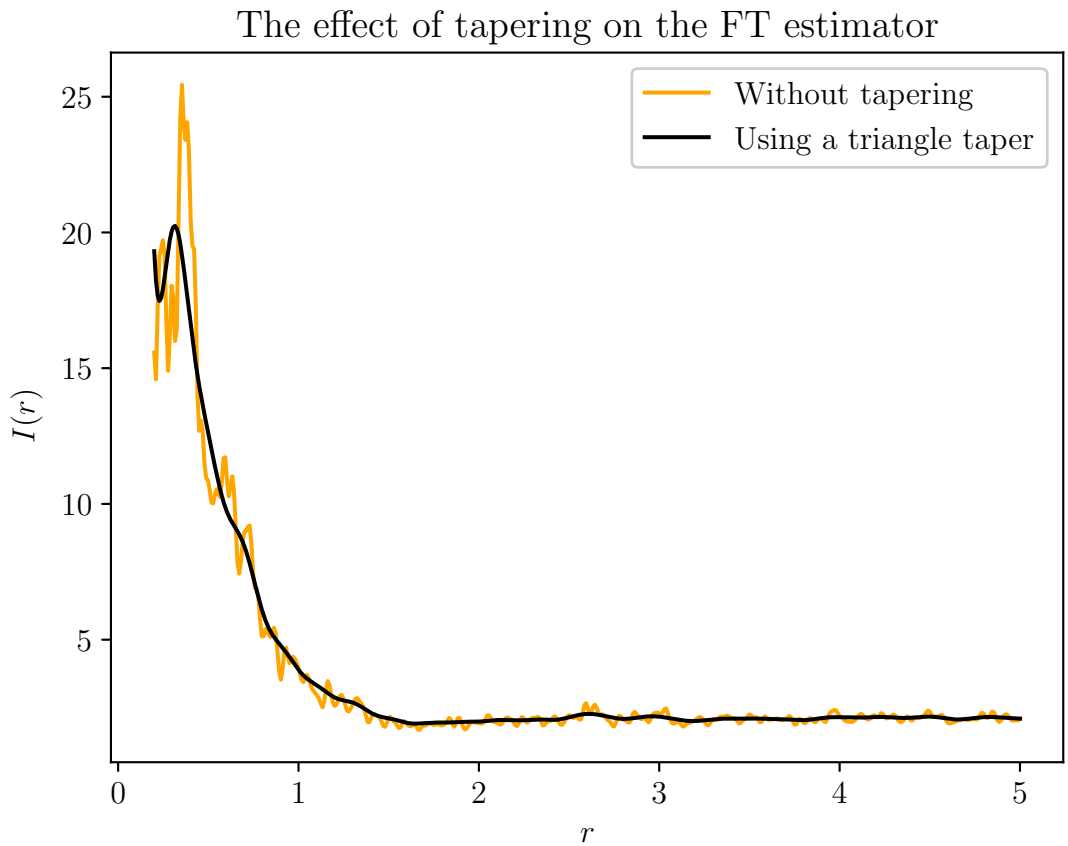


Figure 8: Effect of tapering for a Thomas process of parameters $(0.2, 10, 0.25^2)$

- **Normalized** such that $\|h\|_2 = 1$.

The conditions of isotropy and normalization give $\|h\|_{L^2}^2 = 1$. And by definition:

$$\|h\|_{L^2}^2 = \int_{\mathbb{R}^2} |h(x,y)|^2 \, dx \, dy$$

Switching to polar coordinates:

$$\|h\|_{L^2}^2 = \int_0^{2\pi} \int_{\mathbb{R}^+} |h(r\cos\theta, r\sin\theta)|^2 \, r \, dr \, d\theta$$

As h is isotropic, we have for all θ: $h(r\cos\theta, r\sin\theta) = h(r, 0)$.

As a slight abuse of notation, we'll denote $h(r,0)$ by $h(r)$. Thus:

$$\|h\|_{L^2}^2 = \int_{\mathbb{R}^+} \int_0^{2\pi} |h(r)|^2 \, r \, d\theta \, dr = 2\pi \int_{\mathbb{R}^+} r \, |h(r)|^2 \, dr$$

Therefore, the condition of a valid taper becomes:

$$\boxed{\int_{\mathbb{R}^+} r\,|h(r)|^2\,dr = \frac{1}{2\pi}}$$

All the example functions below are compactly supported, square integrable, and normalized in the L^2 sense :

1. **Triangular taper:**

$$\begin{aligned} h(r) &= \frac{1-\frac{r}{R}}{\sqrt{2\pi\int_0^R\left(1-\frac{r}{R}\right)^2 r\,dr}}\cdot\mathbf{1}_{r\le R} \\ &= \sqrt{\frac{6}{\pi}}\frac{1}{R}\left(1-\frac{r}{R}\right)\cdot\mathbf{1}_{r\le R} \end{aligned}$$

2. **Quartic taper:**

$$\begin{aligned} h(r) &= \frac{\left(1-\left(\frac{r}{R}\right)^2\right)^2}{\sqrt{2\pi\int_0^R\left(1-\left(\frac{r}{R}\right)^2\right)^4 r\,dr}}\cdot\mathbf{1}_{r\le R} \\ &= \sqrt{\frac{5}{\pi}}\frac{1}{R}\left(1-\left(\frac{r}{R}\right)^2\right)^2\cdot\mathbf{1}_{r\le R} \end{aligned}$$

3. **Cosine taper:**

$$\begin{aligned} h(r) &= \frac{\cos\left(\frac{\pi r}{2R}\right)}{\sqrt{2\pi\int_0^R\cos^2\left(\frac{\pi r}{2R}\right) r\,dr}}\cdot\mathbf{1}_{r\le R} \\ &= \sqrt{\frac{2\pi}{\pi^2-4}}\frac{1}{R}\cos\left(\frac{\pi r}{2R}\right)\cdot\mathbf{1}_{r\le R} \end{aligned}$$

4. **Epanechnikov taper:**

$$\begin{aligned} h(r) &= \frac{1-\left(\frac{r}{R}\right)^2}{\sqrt{2\pi\int_0^R\left(1-\left(\frac{r}{R}\right)^2\right)^2 r\,dr}}\cdot\mathbf{1}_{r\le R} \\ &= \sqrt{\frac{3}{\pi}}\frac{1}{R}\left(1-\left(\frac{r}{R}\right)^2\right)\cdot\mathbf{1}_{r\le R} \end{aligned}$$

5. **Bump function:**

$$h(r) = \frac{\exp\left(-\frac{1}{1-\left(\frac{r}{R}\right)^2}\right)}{\sqrt{2\pi\int_0^R\exp\left(-\frac{2}{1-\left(\frac{r}{R}\right)^2}\right) r\,dr}}\cdot\mathbf{1}_{r<R}$$

6. **Gaussian taper (truncated at $R = 3\sigma$):**

$$h(r) = \frac{\exp\left(-\frac{r^2}{2\sigma^2}\right)}{\sqrt{2\pi \int_0^R \exp\left(-\frac{r^2}{\sigma^2}\right) r\, dr}} \cdot \mathbf{1}_{r \le R}$$
$$= \frac{\exp\left(-\frac{r^2}{2\sigma^2}\right)}{\sigma\sqrt{\pi(1 - \exp\left(-\frac{R^2}{\sigma^2}\right))}} \cdot \mathbf{1}_{r \le R},$$

where we choose $R = 3\sigma$ to ensure rapid decay and practical compactness.

These normalized tapers ensure that the squared L^2 energy of the taper is exactly 1, improving the comparability and consistency of spectral estimators across functions. Their compact support also helps mitigate boundary effects when applied to finite observation windows.

Each taper function introduces a different trade-off between **frequency resolution** and **variance smoothing**. Testing multiple tapers allows us to:

- Compare their effectiveness in parameter estimation (bias, variance).
- Select the optimal taper under a given criterion (e.g., minimal contrast error).

To assess the impact of each taper function on the estimation of Thomas process parameters, we implemented a numerical optimization procedure based on the contrast between the empirical and theoretical spectra:

$$\hat{\Delta}(\theta) = \int_\varepsilon^R \left(\hat{f}_h(r) - f_\theta(r)\right)^2 dr,$$

where $f_\theta(r)$ is the theoretical spectral density depending on parameters $\theta = (\nu, \sigma)$ (λ_p was eliminated using the standard estimator $\hat{\lambda}$, see 5.2.2).

Estimates are obtained by minimizing $\hat{\Delta}(\theta)$ using the L-BFGS-B algorithm. Results for each taper are then compared using the squared error with respect to the ground truth parameters.

Results We now apply our estimation pipeline using the six L^2-normalized tapering functions defined previously. The objective is to estimate the parameters ν (mean number of offspring) and σ (dispersion) of a Thomas process that was synthetically generated with ground truth parameters $\nu = 10$ and $\sigma = 0.25$.

The parent intensity λ_p can be deduced from the relationship $\lambda = \lambda_p \cdot \nu$, and since we already have a reliable estimator for λ, we focus on estimating ν and σ by minimizing the contrast function using the Bartlett periodogram with tapering.

In the current case (that of an L^2-normalized taper), the results are very satisfying for the parameter σ, with errors presented in Table 3. However, the error for ν was very high. All tapering functions significantly overestimate the number of offspring, with relative errors exceeding 9500% in the best case. This can be explained by the behavior of the empirical spectral density at small spatial frequencies (small r), where we observe very high values—orders of magnitude above the theoretical curve. These inflated values bias the estimation of ν, which mainly influences the spectral power near the origin.

Taper	$\hat{\sigma}$	Rel. Error (σ)
Triangle	0.2517	0.70%
Quartic	0.2508	0.31%
Cosine	0.2398	4.09%
Bump	0.2281	8.74%
Epanechnikov	0.2366	5.34%
Gaussian	0.2620	4.81%

Table 3: Estimated parameters using L^2-normalized tapering functions

However, starting from a distance of approximately $r = 2$, the empirical spectra stabilize and follow the theoretical Thomas curve quite accurately (except for the Gaussian taper, which exhibits instability over the full domain). As a result, the estimation of σ remains robust, with relative errors below 9% for all tapering functions and less than 1% for the quartic and triangle tapers.

This suggests that tapering helps reduce noise in the mid-to-high frequency range, but fails to correct the sharp bias at low frequencies. Further work could investigate bias correction strategies or alternative weighting schemes in the contrast function.

Alternative approach Given the poor estimation results obtained with L^2-normalized tapering functions, we explored an alternative approach using directly scaled tapers normalized with respect to the observation window area or other natural constants. This change significantly improves the estimation performance, as shown in Table 4 and Fig. 9.

We used the following set of tapering functions, each with compact support and a decay profile suited to emphasize different spatial scales:

1. **Gaussian:**
$$h(r) = \frac{1}{2\pi R^2} \exp\left(-\frac{r^2}{2R^2}\right), \quad \text{with } R = \frac{w}{3},\ r \leq 3R$$

2. **Triangle:**
$$h(r) = \frac{1 - \frac{r}{R}}{wh}, \quad \text{with } R = 0.2w,\ r \leq R$$

3. **Quartic:**

$$h(r) = \frac{(1 - \left(\frac{r}{R}\right)^2)^2}{wh}, \quad R = 0.2w,\ r \leq R$$

4. **Cosine:**

$$h(r) = \frac{\cos\left(\frac{\pi r}{2R}\right)}{wh}, \quad R = 0.2w,\ r \leq R$$

5. **Bump:**

$$h(r) = \frac{\exp\left(-\frac{1}{1-\left(\frac{r}{R}\right)^2}\right)}{wh}, \quad R = 0.2w,\ r < R$$

6. **Epanechnikov:**

$$h(r) = \frac{1 - \left(\frac{r}{R}\right)^2}{wh}, \quad R = 0.2w,\ r \leq R,$$

where w and h are respectively the width and height of our rectangular window.

We first evaluate the performance of each taper function using a fixed initial value $\theta_0 = [15,\ 0.5]$.

Taper	$\hat{\nu}$	$\hat{\sigma}$	Squared Error	Rel. Error (ν)	Rel. Error (σ)
Gaussian	14.57	0.2543	20.89	45.71%	1.73%
Triangle	9.97	0.2517	0.0016	0.39%	0.70%
Quartic	9.89	0.2508	0.0120	1.10%	0.31%
Cosine	9.18	0.2398	0.6646	8.15%	4.09%
Bump	3.91	0.1840	36.99	60.82%	26.39%
Epanechnikov	8.98	0.2366	1.0392	10.19%	5.34%

Table 4: Parameter estimates using fixed $\theta_0 = [15,\ 0.5]$

We then proceed to a grid search over 1000 initializations of $\theta_0 \in [5, 100] \times [0.1, 2.5]$ and report the best results per tapering function (see Table 5).

Taper	Best θ_0	$\hat{\nu}$	$\hat{\sigma}$	Sq. Error	Rel. Error (ν)	Rel. Error (σ)
Gaussian	$[15.21, 0.52]$	14.57	0.2543	20.89	45.71%	1.73%
Triangle	$[16.39, 0.15]$	9.96	0.2517	0.0016	0.39%	0.70%
Quartic	$[14.86, 0.26]$	9.89	0.2508	0.0120	1.10%	0.31%
Cosine	$[9.03, 1.98]$	9.18	0.2398	0.6646	8.15%	4.09%
Bump	$[14.66, 0.49]$	3.93	0.1768	36.81	60.66%	29.27%
Epanechnikov	$[40.73, 1.42]$	10.82	0.2148	0.6731	8.20%	14.06%

Table 5: Best estimates from 1000 starting points using tapers with alternative normalization

The improvement is significant: using this new normalization scheme, the tapering functions produce much more accurate estimates, especially for the triangle and quartic tapers. Even

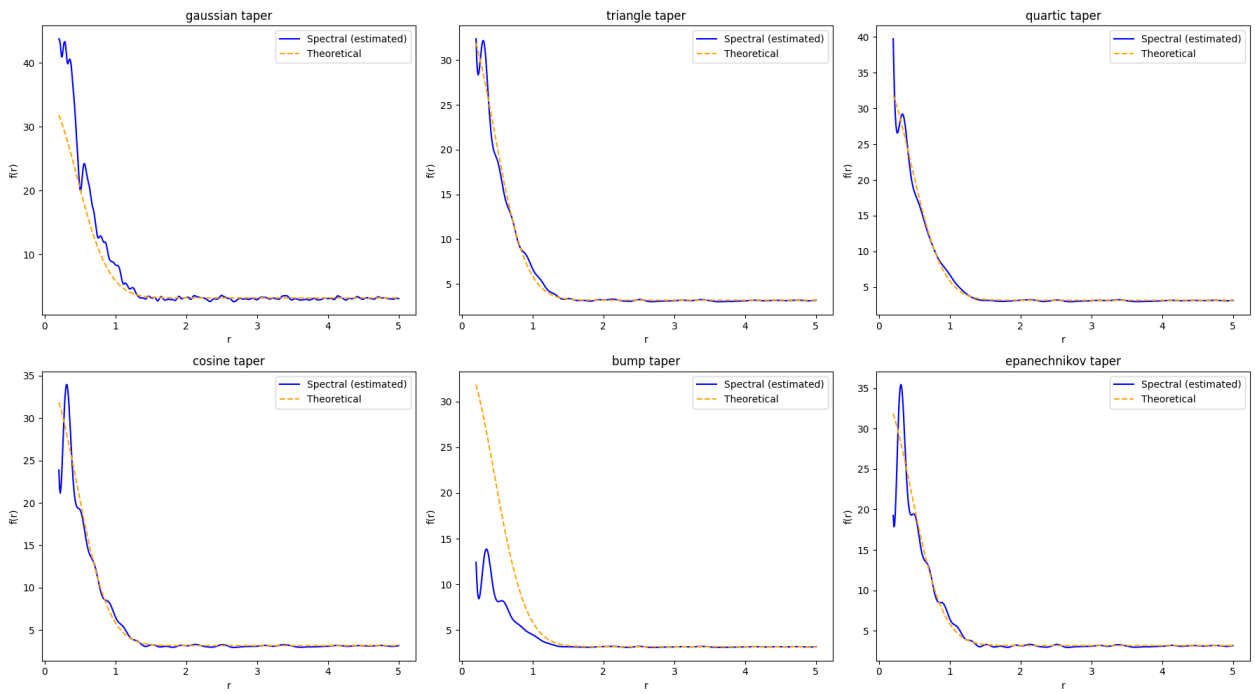


Figure 9: Comparison of theoretical and estimated spectral functions for fixed θ_0

with a fixed initialization, relative errors drop below 1% for ν and σ, and remain under 10% for most other cases. However, we get bad results with a Bump taper; so this taper function can't effecteively reduce edge effects, in our particular case.

These results suggest that the choice of the normalization factor has a major impact on spectral estimation performance. Further work is required to understand the mathematical relationship between the tapering profile, its normalization, and the resulting bias in the estimation of the low-frequency components of the spectrum.

5.3. Application to a superposition of two independent Thomas Processes

5.3.1 Assumptions

Let's assume that we have a realization of a superposition of two independent Thomas point processes with parameters $(\lambda_{p1}, \nu_1, \sigma_1^2)$ and $(\lambda_{p2}, \nu_2, \sigma_2^2)$. We want to estimate the vector of parameters of the superposition $(\lambda_{p1}, \nu_1, \sigma_1^2, \lambda_{p2}, \nu_2, \sigma_2^2)$. However, there are too many parameters (six in total). In order to reduce the number of parameters to make the estimation process easier and quicker, we make some assumptions.

- First, we make the hypothesis that the tree centers are generated by the same Poisson process; which means that $\lambda_{p,1} = \lambda_{p,2} = \lambda_p$.

- Second, we assume that the mean number of offspring of each point process is proportinal to the size of the trees (σ^2). In other words, $\exists \mu \in \mathbb{R}^+ \ \nu_i = \mu\sigma_i^2$. This later hyposthesis is relevant because it means that the larger the tree, the more lIDAR rays are reflected.

5.3.2 Estimation using the PCF

Using the assumptions of the previous part, the pair correlation function of the superposition is given by the following formula, the proof of which can be found in Appendix E:

$$g(r) = 1 + \frac{1}{4\pi\lambda_p}\left(\frac{\sigma_1^2}{(\sigma_1^2+\sigma_2^2)^2}\exp\left(\frac{-r^2}{4\sigma_1^2}\right) + \frac{\sigma_2^2}{(\sigma_1^2+\sigma_2^2)^2}\exp\left(\frac{-r^2}{4\sigma_2^2}\right)\right).$$

This function depends on the vector of parameters $(\lambda_p, \sigma_1^2, \sigma_2^2)$ that we estimate first. Then, in order to estimate the remaining parameters, we first estimate the global intensity $\hat{\lambda}$, then, we estimate the individual intensities, proven in Appendix D:

$$\lambda_1 = \frac{\sigma_1^2}{\sigma_1^2+\sigma_2^2}\lambda$$
$$\lambda_2 = \frac{\sigma_2^2}{\sigma_1^2+\sigma_2^2}\lambda.$$

Then, we can deduce estimates for λ_1 and λ_2: $\hat{\lambda}_1 = \frac{\sigma_1^2}{\sigma_1^2+\sigma_2^2}\hat{\lambda}$ and $\hat{\lambda}_2 = \frac{\sigma_2^2}{\sigma_1^2+\sigma_2^2}\hat{\lambda}$, as well as $\hat{\nu}_1$ and $\hat{\nu}_2$: $\hat{\nu}_1 = \frac{\hat{\lambda}_1}{\hat{\lambda}_p}, \hat{\nu}_2 = \frac{\hat{\lambda}_2}{\hat{\lambda}_p}$

Applying this procedure to a superposition of point processes, we get the results in Table 6.

Parameter	λ_p	ν_1	σ_1	ν_2	σ_2
True values	0.25	9.00	0.3	20.25	0.45
Estimated values	0.11	34.64	0.43	34.64	0.43
Percentage error	56%	285%	43%	71%	4.4%

Table 6: Parameter estimates for a superposition of Thomas processes using the PCF

The estimation is not satisfactory. The percentage errors are too large. In particular, the estimated value of σ_1 and σ_2 are the same, which is normal giving the fact that the expression of g(r) is symetric according to σ_1 and σ_2. We'll mitigate this problem in the following section.

5.3.3 Estimation using the Fourier Transform

In this section, we estimate the parameters of a superposition of two independent Thomas processes using a Tapered Bartlett Periodogram. The Fourier Transform of a superposition

Parameter	μ	λ_p	σ_1	σ_2	ν_1	ν_2
True values	100.00	0.100	0.25	0.45	6.25	20.25
Estimated values	102.17	0.111	0.24	0.44	5.79	19.89
Percentage error	2.17%	11.37%	4.78%	1.35%	7.37%	1.78%

Table 7: Parameter estimates for a superposition of Thomas processes using the FT

of two point processes, as proven in Appendix C, is:

$$\begin{aligned} f(\|k\|_2^2) &= f_1(\|k\|_2^2) + f_2(\|k\|_2^2) \\ &= \lambda_1 \left(1 + \nu_1 \exp\left(-4\pi^2\sigma_1^2 \|k\|_2^2\right)\right) + \lambda_2 \left(1 + \nu_2 \exp\left(-4\pi^2\sigma_2^2 \|k\|_2^2\right)\right) \end{aligned}$$

The following way to rewrite f, mitigate the identifiability problem by introducing $\hat{\lambda}$, the estimator of the global intensity, and also breaks the symmetry by using σ for the first process but ν for the second:

$$f(\|k\|_2^2) = \lambda_p(\mu\sigma_1^2 + \nu_2) + \frac{\hat{\lambda}}{\mu\sigma_1^2 + \nu_2} \left(\mu^2\sigma_1^4 e^{-4\pi^2\sigma_1^2\|k\|_2^2} + \nu_2^2 e^{-4\pi^2 \frac{\nu_2}{\mu}\|k\|_2^2}\right).$$

We use the triangle taper introduced in 2, estimate $(\hat{\mu}, \hat{\lambda_p}, \hat{\sigma_1}, \hat{\nu_2})$ deduce $\hat{\sigma_2}$ and $\hat{\nu_1}$. We also run the estimation for a grid of initial parameter values and retain the best estimation based on RMSE. The size of the window was slightly increased, to increase the precision of the estimation. Table 7 summarizes our results.

6. Conclusion and Perspectives

6.1. Conclusion

In conclusion, in this project, our aim was to harness spatial point processes, particularly the Thomas process and superpositions of independant Thomas processes, to model and estimate the spatial distribution of trees from LiDAR data.

We implemented parameter estimation techniques based on the pair correlation function and the Fourier transform. For a single Thomas process, both methods produced accurate estimates for the parameters. Tapering further improved the robustness of Fourier-based spectral estimation. However, when dealing with the superposition of multiple Thomas processes, the estimation became significantly more challenging. Despite applying simplifying assumptions and identifiability strategies, we observed high estimation errors, particularly for the PCF, as the formula was symetric and didn't include the parameter μ. However, breaking the symmetry in the Fourier Transform formula allowed us to get significantly better results.

6.2. Perspectives

As a perspective, a future work for the project would be to Several directions could be explored in future work. First, future efforts should aim to better understand and mitigate the identifiability problems observed when estimating parameters in the superposition of Thomas processes. Bayesian approaches with informative priors, or the introduction of constraints, could help disentangle overlapping effects. Secondly, While our methodology worked on one Thomas process, and on the superposition of two Thomas processes, extending tapering-based spectral estimation to a more general framework of a superposition of many independent processes remains a big challenge. Finally, combining this statistical approach with machine learning and deep learning could yield hybrid methods that leverage both theoretical rigor and data-driven power.

Once we have a method that performs well on a superposition of point processes, the next step is to apply it to real LiDAR data. This will allow us to assess its practical effectiveness and evaluate its robustness in the presence of real-world noise and irregularities.

Declaration of the use of AI-assisted technologies in the writing process
Some parts of this report were drafted or revised with the assistance of generative AI technologies (mainly ChatGPT). All content has been reviewed and verified by the authors, and we take full responsibility for the content of the current report.

References

[1] Akira S Mori, Kenneth P Lertzman, and Lena Gustafsson. Biodiversity and ecosystem services in forest ecosystems: a research agenda for applied forest ecology. *Journal of Applied Ecology*, 54(1):12–27, 2017.

[2] Georg Winkel, Glenn Galloway, Carol J Pierce Colfer, Wil de Jong, Pia Katila, and Pablo Pacheco. The impacts of the sustainable development goals on forests and people–conclusions and the way forward. *Sustainable development goals: Their impacts on forests and people*, pages 601–617, 2019.

[3] Raghu Raman, Sudheesh Manalil, Dávid Lóránt Dénes, and Prema Nedungadi. The role of forestry sciences in combating climate change and advancing sustainable development goals. *Frontiers in Forests and Global Change*, 7:1409667, 2024.

[4] Bohdan Konôpka, Vladimír Šebeň, and Jozef Pajtík. Species composition and carbon stock of tree cover at a postdisturbance area in tatra national park, western carpathians. *Mountain Research and Development*, 39(1):R71–R80, 2019.

[5] Leilei Yang, Junhui Zhang, Jiahui Wang, Yue Gu, and Shijie Han. A linear positive relationship between tree species diversity and forest productivity across forest-dominated natural reserves on a large spatial scale. *Forest Ecology and Management*, 548:121409, 2023.

[6] Fabian Ewald Fassnacht, Joanne C White, Michael A Wulder, and Erik Næsset. Remote sensing in forestry: current challenges, considerations and directions. *Forestry: An International Journal of Forest Research*, 97(1):11–37, 2024.

[7] Ramón Alberto Díaz-Varela and Eduardo González-Ferreiro. 3d point clouds in forest remote sensing, 2021.

[8] Xinlian Liang, Juha Hyyppä, Harri Kaartinen, Matti Lehtomäki, Jiri Pyörälä, Norbert Pfeifer, Markus Holopainen, Gábor Brolly, Pirotti Francesco, Jan Hackenberg, et al. International benchmarking of terrestrial laser scanning approaches for forest inventories. *ISPRS journal of photogrammetry and remote sensing*, 144:137–179, 2018.

[9] Harri Kaartinen, Juha Hyyppä, Xiaowei Yu, Mikko Vastaranta, Hannu Hyyppä, Antero Kukko, Markus Holopainen, Christian Heipke, Manuela Hirschmugl, Felix Morsdorf, et al. An international comparison of individual tree detection and extraction using airborne laser scanning. *Remote Sensing*, 4(4):950–974, 2012.

[10] Lothar Eysn, Markus Hollaus, Eva Lindberg, Frédéric Berger, Jean-Matthieu Monnet, Michele Dalponte, Milan Kobal, Marco Pellegrini, Emanuele Lingua, Domen Mongus, et al. A benchmark of lidar-based single tree detection methods using heterogeneous forest data from the alpine space. *Forests*, 6(5):1721–1747, 2015.

[11] Yunsheng Wang, Juha Hyyppä, Xinlian Liang, Harri Kaartinen, Xiaowei Yu, Eva Lindberg, Johan Holmgren, Yuchu Qin, Clément Mallet, António Ferraz, et al. International benchmarking of the individual tree detection methods for modeling 3-d canopy structure for silviculture and forest ecology using airborne laser scanning. *IEEE Transactions on Geoscience and Remote Sensing*, 54(9):5011–5027, 2016.

[12] Tuomas A. Rajala, Sofia C. Olhede, Jake P. Grainger, and David J. Murrell. *What is the Fourier Transform of a Spatial Point Process?* IEEE TRANSACTIONS ON INFORMATION THEORY, 2023.

[13] Adrian Baddeley. *Spatial Point Processes and their Applications*. School of Mathematics & Statistics, University of Western Australia, 2007.

[14] Sung Nok Chiu, Dietrich Stoyan, Wilfrid S Kendall, and Joseph Mecke. *Stochastic geometry and its applications*. John Wiley & Sons, 2013.

[15] Janine Illian, Antti Penttinen, Helga Stoyan, and Dietrich Stoyan. *Statistical analysis and modelling of spatial point patterns*. John Wiley & Sons, 2008.

Appendix

A. Intensity of a superposition of point processes

For a Borel set B,

$$\begin{aligned}
\mathbb{E}[\Phi(B)] &= \mathbb{E}[\Phi_1(B) + \Phi_2(B)] \\
&= \mathbb{E}[\Phi_1(B)] + \mathbb{E}[\Phi_2(B)] \\
&= \int_B \lambda_1(x)\,dx + \int_B \lambda_2(x)\,dx \\
&= \int_B (\lambda_1(x) + \lambda_2(x))\,dx
\end{aligned}$$

But, by definition, $\mathbb{E}[\Phi(B)] = \int_B \lambda(x)\,dx$

Thus, $\lambda = \lambda_1 + \lambda_2$

B. Pair correlation function of a superposition of point processes

For two stationary isotropic point processes with pair correlation functions g_1 and g_2, the pair correlation function of the superposition is:

$$\begin{aligned}
g(r) &= \frac{\rho^2(r)}{\lambda^2} \\
&= \frac{\rho_1^2(r) + \rho_2^2(r) + 2\lambda_1\lambda_2}{(\lambda_1 + \lambda_2)^2} \\
&= \frac{\lambda_1^2 g_1(r) + \lambda_2^2 g_2(r) + 2\lambda_1\lambda_2}{(\lambda_1 + \lambda_2)^2}
\end{aligned}$$

C. Fourier Transform of a superposition of point processes

For two isotropic point processes with Fourier transforms f_1 and f_2, the Fourier Transform of the superposition is:

$$
\begin{aligned}
f(k) &= \lambda + \lambda^2 \int_{\mathbb{R}^d} e^{-2i\pi k\cdot z} \left(g(z) - 1\right)\, dz \\
&= \lambda + \int_{\mathbb{R}^d} e^{-2i\pi k\cdot z} \left(\lambda^2 g(z) - \lambda^2\right)\, dz \\
&= \lambda_1 + \lambda_2 + \int_{\mathbb{R}^d} e^{-2i\pi k\cdot z} \left(\lambda_1^2 g_1(z) + \lambda_2^2 g_2(z) + 2\lambda_1\lambda_2 - (\lambda_1 + \lambda_2)^2\right)\, dz \\
&= \lambda_1 + \lambda_2 + \int_{\mathbb{R}^d} e^{-2i\pi k\cdot z} \left(\lambda_1^2 g_1(z) + \lambda_2^2 g_2(z) - \lambda_1^2 - \lambda_2^2\right)\, dz \\
&= \lambda_1 + \lambda_2 + \int_{\mathbb{R}^d} e^{-2i\pi k\cdot z} \left(\lambda_1^2 \left[g_1(z) - 1\right] + \lambda_2^2 \left[g_2(z) - 1\right]\right)\, dz \\
&= (\lambda_1 + \lambda_1^2 \int_{\mathbb{R}^d} e^{-2i\pi k\cdot z} \left(g_1(z) - 1\right)\, dz) + (\lambda_2 + \lambda_2^2 \int_{\mathbb{R}^d} e^{-2i\pi k\cdot z} \left(g_2(z) - 1\right)\, dz) \\
&= f_1(k) + f_2(k)
\end{aligned}
$$

Then, by induction, we can show that in general, the Fourier Transform of a superposition of mutually independent point Processes is the sum of the Fourier Transform of each point process.

D. Expression of the individual intensities for a superposition

Consider two point processes satisfying tha same hypotheses as in section 5.3.1. Then, $\lambda_{p,1} = \lambda_{p,2} = \lambda_p$ and $\nu_i = \mu\sigma_i^2$, for $i = 1, 2$. By multiplying the later equality by λ_p, we get $\lambda_i = \lambda_p \mu \sigma_i^2$. So, $\frac{\lambda_1}{\lambda_2} = \frac{\sigma_1^2}{\sigma_2^2}$. In addition, we know that $\lambda = \lambda_1 + \lambda_2$.

We get the following equations:

$$
\begin{cases}
\lambda = \lambda_1 + \lambda_2 \\
\lambda_1 = \lambda_2 \frac{\sigma_1^2}{\sigma_2^2}
\end{cases}
$$

So, $\lambda_1 = \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2}\lambda$ and $\lambda_2 = \frac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2}\lambda$.

E. PCF of a superposition under hypotheses 5.3.1

Consider two independent Thomas processes satisfying tha same hypotheses as in section 5.3.1. We know that for the superposition:

$$
g(r) = \frac{\lambda_1^2 g_1(r) + \lambda_2^2 g_2(r) + 2\lambda_1\lambda_2}{\lambda^2}
$$

Using the result of the previous section, we get

$$g(r) = \frac{a_1^2\lambda^2 g_1(r) + a_2^2\lambda^2 g_2(r) + 2a_1a_2\lambda^2}{\lambda^2},$$

with $a_1 = \frac{\sigma_1^2}{\sigma_1^2+\sigma_2^2}$ and $a_2 = \frac{\sigma_2^2}{\sigma_1^2+\sigma_2^2}$.

Hence,

$$\begin{aligned}
g(r) &= a_1^2 g_1(r) + a_2^2 g_2(r) + 2a_1a_2 \\
&= a_1^2(1 + \frac{1}{4\pi\lambda_p}\exp(\frac{-r^2}{4\sigma_1^2})) + a_2^2(1 + \frac{1}{4\pi\lambda_p}\exp(\frac{-r^2}{4\sigma_2^2})) + 2a_1a_2 \\
&= \frac{1}{4\pi\lambda_p}(a_1^2\exp(\frac{-r^2}{4\sigma_1^2}) + a_2^2\exp(\frac{-r^2}{4\sigma_2^2})) + a_1^2 + a_2^2 + 2a_1a_2 \\
&= \frac{1}{4\pi\lambda_p}(a_1^2\exp(\frac{-r^2}{4\sigma_1^2}) + a_2^2\exp(\frac{-r^2}{4\sigma_2^2})) + (a_1+a_2)^2 \\
&= \frac{1}{4\pi\lambda_p}(a_1^2\exp(\frac{-r^2}{4\sigma_1^2}) + a_2^2\exp(\frac{-r^2}{4\sigma_2^2})) + 1
\end{aligned}$$

F. Intensity estimation

We consider two estimators for the intensity λ of a spatial point process.

1. Standard Estimator

The classical estimator is defined as:

$$\hat{\lambda} = \frac{N(W)}{\nu(W)},$$

where $N(W)$ denotes the number of observed points in the observation window W, and $\nu(W)$ is the Lebesgue measure (i.e., the area in 2D) of the window.

Figure 10 shows the convergence of this estimator as the window area grows.

2. Surface-Based Estimator

An alternative estimator, denoted $\hat{\lambda}_S(r)$, is defined by:

$$\hat{\lambda}_S(r) = \sum_{x\in N} p_S(x,r),$$

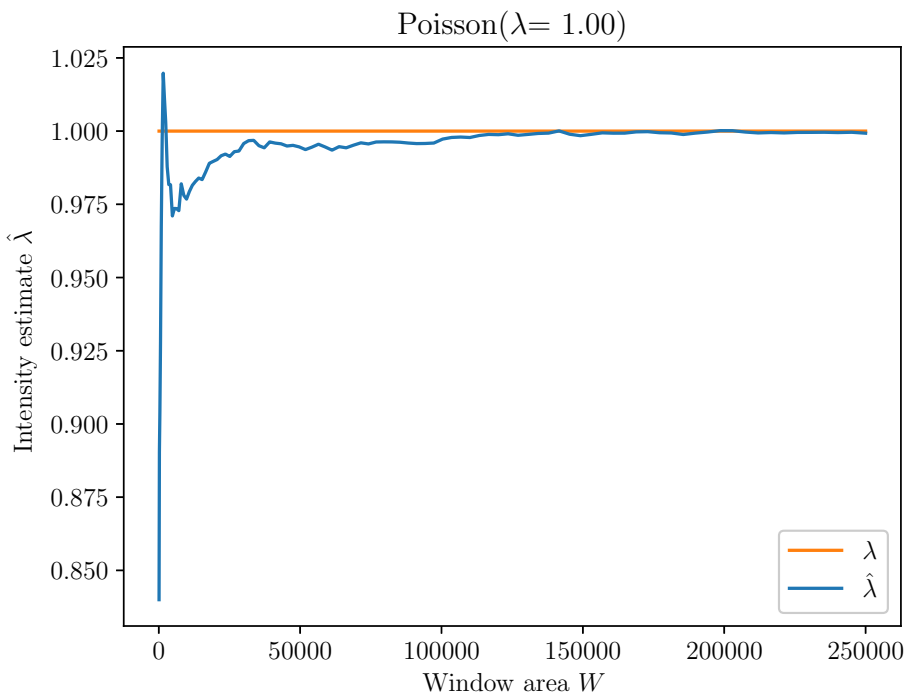


Figure 10: Convergence of $\hat{\lambda}$ as the window area grows

where the contribution of each point x is given by the function:

$$p_S(x,r) = \frac{\nu_{d-1}\left(W \cap \partial b(x,r)\right)}{d b_d r^{d-1}\,\overline{\gamma}_W(r)}.$$

Here:

- $b(x,r)$ denotes the ball of radius r centered at x,
- $\partial b(x,r)$ is its boundary (i.e., the circle of radius r),
- ν_{d-1} is the $(d-1)$-dimensional Hausdorff measure (length in 2D),
- b_d is the volume of the unit ball in d dimensions,
- $\overline{\gamma}_W(r)$ is a normalizing constant that accounts for edge effects.

In the two-dimensional rectangular case with window dimensions $w \times h$, this expression simplifies to:

$$p_S(x,r) = \frac{\nu_1\left(W \cap \partial b(x,r)\right)}{2\pi r\,\overline{\gamma}_W(r)},$$

with the correction term $\overline{\gamma}_W(r)$ given by:

$$\overline{\gamma}_W(r) = \begin{cases} wh - \dfrac{2r}{\pi}(w+h) + \dfrac{r^2}{\pi}, & \text{if } r \leq w, \\ \dfrac{wh}{\pi}\left(2\arcsin\left(\dfrac{w}{r}\right) - \dfrac{w}{h} - 2\left(\dfrac{r}{w} - \sqrt{\dfrac{r^2}{w^2} - 1}\right)\right), & \text{if } w < r \leq h. \end{cases}$$

The value of $\nu_1(W \cap \partial b(x,r))$ is computed numerically using the `rect` function from the module `src/point_process.py`. This function returns a vector `ra` such that:

- `ra[0]` corresponds to $\nu(W \cap b(x,r))$,
- `ra[1]` corresponds to $\nu_1(W \cap \partial b(x,r))$.

This estimator is particularly useful when seeking to mitigate boundary effects or to localize estimation using information on the interaction of the observation window with circular boundaries around observed points. As a matter of fact, it's the one used in the PCF estimator we use.

Contact Information

For correspondence, please contact:

- **Fred Jiagho**
  Email: `fred.jiagho-tsabou@student-cs.fr`
- **Rajvi Shah**
  Email: `rajvi.shah@student-cs.fr`
- **Oscar Goudon de Lalande**
  Email: `oscar.goudon-de-lalande@student-cs.fr`
- **Mouad Leachouri**
  Email: `mouad.leachouri@student-cs.fr`
- **Chihab Majdolin**
  Email: `chihab.majdolin@student-cs.fr`